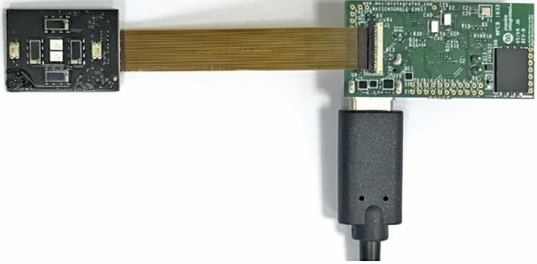


Interesting Reference Designs Of PULSE OXIMETER

Title of Reference Design	Pulse Oximeter	Pulse Oximeter And Heart-Rate Monitor
Image		
Key Attractions	The reference design supports signal processing, wireless connectivity, and power management. It integrates an external display controller, an analogue front end, and data storage.	The reference design collects biometric signals, supports multiple interfaces, and enables data acquisition with low power use. It integrates an optical sensor and rechargeable battery.
Highlights	The Renesas pulse oximeter reference design is based on the RL78/G13 16-bit microcontroller but requires additional external hardware to meet specific requirements. Since the microcontroller lacks an integrated display controller, an external LCD display controller is added, and the DAC for operating the LED driver is implemented externally. Wireless connectivity can be enabled by connecting a Zigbee or Bluetooth module via the CSI port. The RL78/G13 includes a hardware MAC function for signal processing algorithms that reduce artefacts in SpO2 measurements and features on-chip data flash for implementing the look-up table functionality. The microcontroller is suited for applications in ambulatory and hospital markets when combined with an external analogue front end, including a 12-bit ADC. The firmware manages LED and IR-LED control, signal acquisition, signal processing, LCD display output, and SpO2 data storage in the data flash. Power management software is necessary for low power consumption, and a communication driver can enable USB or wireless connectivity with other devices.	The MAXREFDES280 reference design integrates into a wrist-worn device to collect biometric signals, streaming raw photoplethysmography (PPG) data for monitoring vital signs like heart health and blood oxygenation. It provides a platform to evaluate the MAX86171 optical analogue front-end across body sites, focusing on wrist applications. Supporting both I2C and SPI interfaces, the MAX86171 features two optical readout channels that operate simultaneously for data acquisition. Users can configure the band to optimise signal quality while maintaining low power consumption, making it suitable for wearable designs. The hardware setup includes two boards: the MAXSensorBLE# for data acquisition and the MAX86171_USB# as the sensor daughter board. The system runs on a lithium polymer battery, recharged via a USB-C port. The band includes the MAX86171ENI+ in a 28-bump wafer-level package.
Applications	The reference design is suitable for medical applications in various settings.	The reference design is suitable for wearable health monitoring.
OEM Brand	Renesas	Analog Devices (ADI)
URL	https://www.electronicsforu.com/electronics-projects/pulse-oximeter-reference-design	https://www.electronicsforu.com/electronics-projects/pulse-oximeter-and-heart-rate-monitor-reference-design